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ESP32 Defence Drone

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MODULE 01.1

INDIAN MILITARY FLEET: STRATEGIC & MALE SURVEILLANCE UAVS

01 — IAI HERON / HERON MK II

MEDIUM-ALTITUDE LONG-ENDURANCE
IAI Heron / Heron Mk II Army & IAF
Primary eye in the sky along the Line of Actual Control (LAC) in Eastern Ladakh and high-altitude border posts.
Endurance: 45+ Hours
Ceiling: 35,000 Ft
Payload: SATCOM + SAR
Range: Beyond-Line-of-Sight

02 — MQ-9B Predator

HIGH-ALTITUDE LONG-ENDURANCE
MQ-9B Predator Indian Navy
Operated across the Indian Ocean Region for maritime domain awareness, anti-submarine warfare cueing, and surface search.
Endurance: 35+ Hours
Ceiling: 40,000+ Ft
Payload: Maritime 360° Radar
Role: Wide-Area Maritime ISR

03 — HERMES 900 / DRISHTI-10

INDIGENOUS INDUCTION (DRISHTI-10)
Hermes 900 / Drishti-10
Navy & Army
All-weather advanced multi-payload platform manufactured under Make in India for round-the-clock coastal and land border patrols.
Endurance: 36 Hours
Payload Cap: 450 kg
Sensors: EO/IR + EW Jammer

STRATEGIC FLEET ROLE: Persistent border observation, deep electronic intercepts, maritime chokepoint security
STATUS: ACTIVE IN ARMED FORCES REGIMENTS

MODULE 01.2

INDIAN DEFENCE FLEET: MICRO, NANO & URBAN RECONNAISSANCE

01 — FLIR BLACK HORNET NANO

NSG & PARA SF
POCKET-SIZED HELICOPTER DRONE DEPLOYED BY SPECIAL FORCES FOR ROOM-CLEARING, HOSTAGE RESCUE, AND COUNTER-TERROR INDOOR OPERATIONS.
WEIGHT: 33 GRAMS
FLIGHT TIME: 25 MINS
ACOUSTIC SIGN.: NEAR-SILENT
OPTICS: THERMAL + HD DAY

02 — IDEAFORGE NETRA SERIES

ARMY, CRPF, BSF
WORKHORSE TACTICAL QUADROTOR FOR FORWARD BASE SECURITY, BORDER INFILTRATION TRACKING, AND COUNTER-INSURGENCY OPERATIONS IN J&K.
RANGE: 5–10 KM
ENDURANCE: 60–90 MINS
PAYLOAD: 10X OPTICAL ZOOM
ALTITUDE: HIGH HIMALAYAN READY

03 — MICRO-TACTICAL QUADROTORS

URBAN WARFARE UNITS
ULTRA-COMPACT QUADROTORS ENGINEERED FOR RAPID DEPLOYMENT INSIDE TUNNELS, BUNKERS, AND CONFINED URBAN SETTINGS WITHOUT EXTERNAL GPS.
CLASS: SUB-250G MICRO UAV
NAVIGATION: OPTICAL FLOW & IMU
PAYLOAD: M-LF/RF SNIFFING
ADVANTAGE: ZERO RADAR FOOTPRINT

PROJECT CONNECTION: THE SUB-150G CLASS IS THE IDEAL FOUNDATION FOR THE ESP32 MICRO-QUADROTOR PLATFORM DEVELOPED IN THIS RESEARCH.

MODULE 01.3

INDIAN DEFENCE FLEET: TACTICAL STRIKE & LOITERING MUNITIONS

01 — IAI HAROP (KAMIKAZE UAV)

ANTI-RADIATION / KAMIKAZE
IAI HAROP (KAMIKAZE UAV)
IAF PRECISION STRIKE
LOITERS OVER HOSTILE AIRSPACE TO DETECT AND HOME IN ON ENEMY RADAR EMISSIONS, CRASHING DIRECTLY INTO SAM BATTERIES.
WARHEAD: 23 KG HE
RANGE: 1,000 KM
LOITER TIME: 6 HOURS
GUIDANCE: PASSIVE RF HOMING

02 — NAGASTRA-1 (SOLAR INDUSTRIES)

MAN-PORTABLE TACTICAL
NAGASTRA-1 (SOLAR INDUSTRIES)
INDIAN ARMY INFANTRY
INDIGENOUSLY DEVELOPED LIGHTWEIGHT SUICIDE DRONE DESIGNED TO CONDUCT GPS-GUIDED SURGICAL STRIKES ON TERRORIST HIDEOUTS AND TRENCHES.
WEIGHT: 9 KG (MAN-PORTABLE)
LOITER TIME: 60 MINS
WARHEAD: 1–2 KG FRAGMENTATION
STRIKE ACCURACY: < 2 METERS

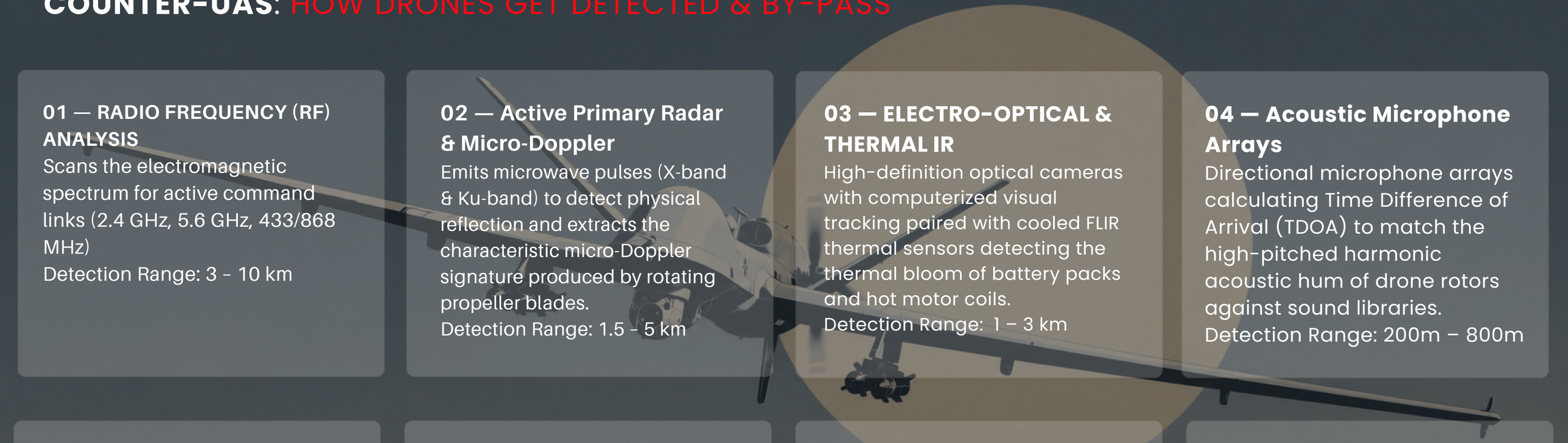
03 — WARMATE & ALFA-S SWARMS

SWARM STRIKE CAPABILITY
WARMATE & ALFA-S SWARMS
SPECIAL FORCES & ARMY
MICRO-LOITERING PLATFORMS CAPABLE OF OPERATING IN SYNCHRONIZED AI-DRIVEN SWARMS TO OVERWHELM ADVERSARY AIR DEFENSE NETWORKS.
SWARM SIZE: 20–50 UNITS
RANGE: 30–50 KM
PAYLOAD: HEAT / THERMOBARIC
NETWORK: MESH AI RELAY

MISSION PROFILE: SEAD (SUPPRESSION OF ENEMY AIR DEFENSES), SURGICAL DECAPITATION STRIKES, ANTI-ARMOR SUPPRESSION
PARADIGM: LOW-COST HIGH-PRECISION ATTRITION

MODULE 02.1

COUNTER-UAS: HOW DRONES GET DETECTED & BY-PASS



01 — RADIO FREQUENCY (RF) ANALYSIS

Scans the electromagnetic spectrum for active command links (2.4 GHz, 5.6 GHz, 433/868 MHz)
Detection Range: 3 - 10 km

02 — Active Primary Radar & Micro-Doppler

Emits microwave pulses (X-band & Ku-band) to detect physical reflection and extracts the characteristic micro-Doppler signature produced by rotating propeller blades.
Detection Range: 1.5 - 5 km

03 — ELECTRO-OPTICAL & THERMAL IR

High-definition optical cameras with computerized visual tracking paired with cooled FLIR thermal sensors detecting the thermal bloom of battery packs and hot motor coils.
Detection Range: 1 – 3 km

04 — Acoustic Microphone Arrays

Directional microphone arrays calculating Time Difference of Arrival (TDOA) to match the high-pitched harmonic acoustic hum of drone rotors against sound libraries.
Detection Range: 200m – 800m

01 / RF EMISSION

Passive Radio Silence & Autonomous Waypoints
Operating in strict RX-only or zero-transmission mode. Pre-programmed autonomous flight paths eliminate the ground-to-drone RF telemetry that RF scanners trigger on.

02 / GPS SPOOFING IMMUNITY

Inertial Dead-Reckoning (MPU6050)
Eliminates dependence on vulnerable civilian GPS frequencies. Stabilized via on-board 6-axis IMU accelerometer and gyroscope rate integration to defeat GPS jammers.

03 / RADAR EVASION

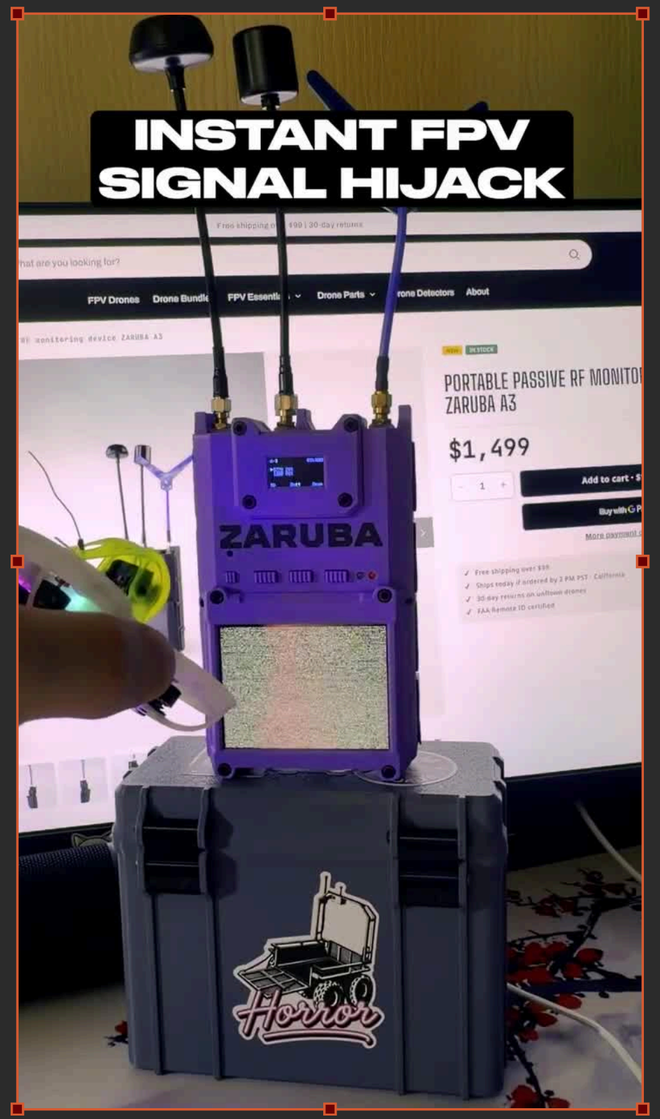
Micro-Scale Architecture & Polymers
Sub-100mm carbon fiber/polymer structure creates an RCS below 0.001 m^2. Military radar algorithms filter this out as atmospheric clutter or bird movement.

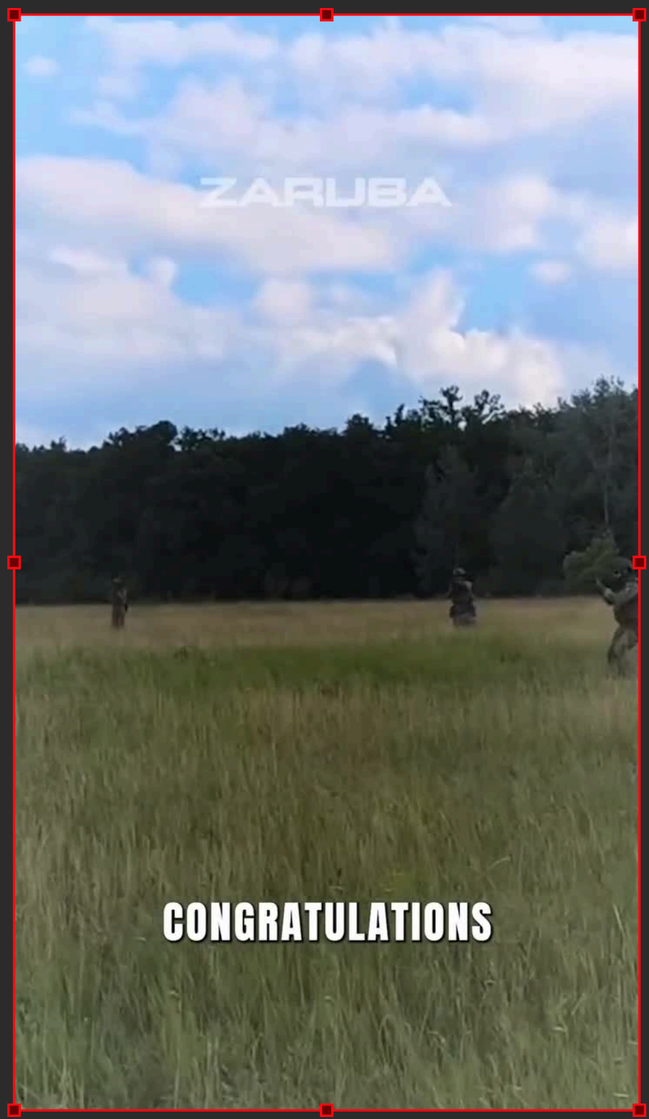
04 / ACOUSTIC DISSIPATION

8520 Coreless Brushed Drive
High-efficiency micro-rotors operating with low tip-speed noise signatures. Sound dissipates into ambient ground background noise at altitudes above 15 meters.

MODULE 02.2

DRONE DETECTOR MACHINERIES: ZARUBA DRONE DETECTOR





MODULE 03.1

ESP32 DRONE COMPONENTS: LABELLED SUBSYSTEMS ARCHITECTURE

MPU6050 6-AXIS IMU
I²C BUS
Measures roll, pitch, yaw angular velocity & linear acceleration (SDA: GPIO21, SCL: GPIO22).

ESP8266 PAYLOAD
INDEPENDENT
Dedicated wireless surveillance payload processor with isolated battery & antenna.

CENTRAL CONTROLLER
ESP32
ESP32 FLIGHT CONTROLLER CORE
ESP8266 PAYLOAD BRACKET

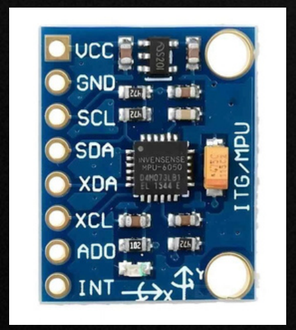
POWER / ELECTRICAL
1S 3.7V LiPo Input
5V Step-Up Boost Rail
470µF Bulk Filtering

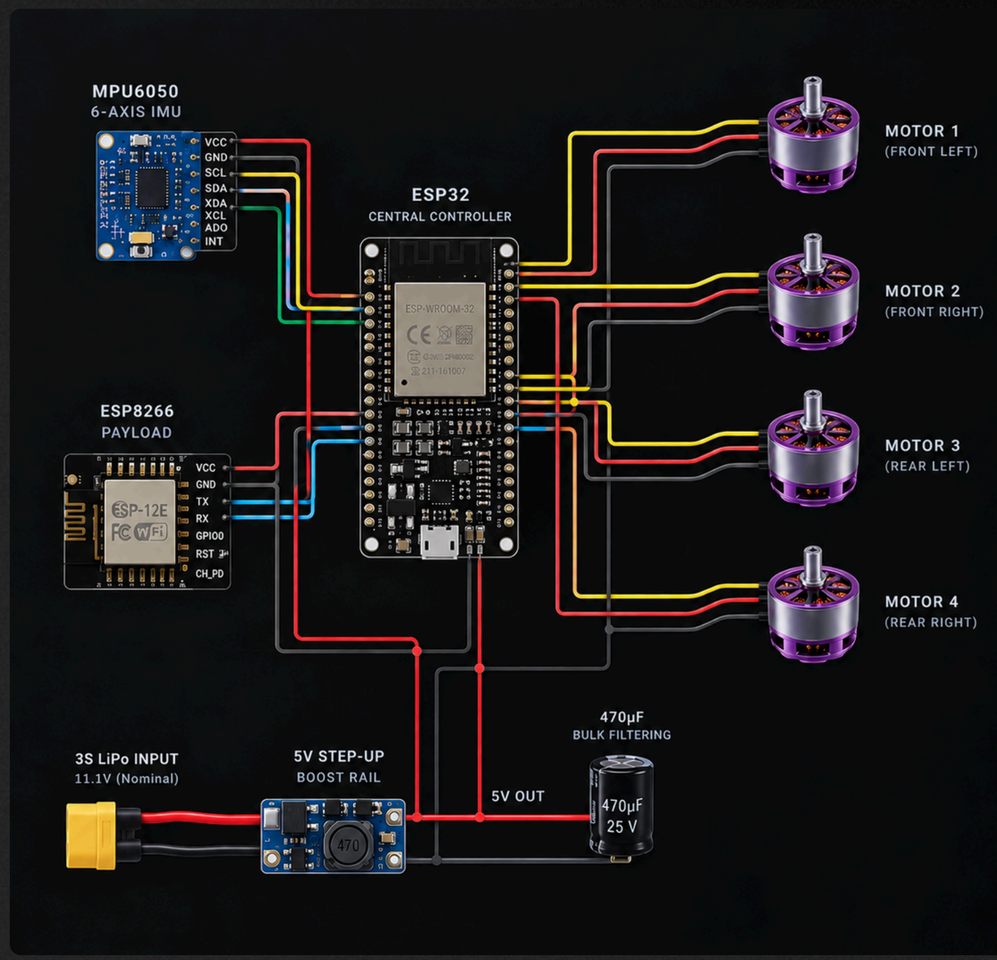
MOTOR 1 (FRONT LEFT)
CCW ↻
8520 Coreless Brushed DC Motor driven by GPIO25 PWM via A2SHB MOSFET.

MOTOR 2 (FRONT RIGHT)
CW ↻
8520 Coreless Brushed DC Motor driven by GPIO25 PWM via A2SHB MOSFET.

MOTOR 3 (REAR LEFT)
CW ↻
Clockwise torque compensation rotor driven by GPIO27 PWM stage.

MOTOR 4 (REAR RIGHT)
CCW ↻
Counter-clockwise rotor driven by GPIO14 PWM completing balanced yaw moment.





MODULE 03.2

WORKING PRINCIPLE: AVIONICS FEEDBACK & PID FLIGHT LOOP

STAGE 01 — INERTIAL SENSING

MPU6050 samples 3-axis gyro and 3-axis accelerometer registers at 200–400 Hz over the I²C bus.
Data: Roll/Pitch/Yaw rates + g-vectors

STAGE 02 — SENSOR FUSION

ESP32 runs a Complementary / Kalman filter combining gyro high-rate response with accel gravity vector drift correction.
Output: True spatial Euler angles (θ, φ, ψ)

STAGE 03 — PID ALGORITHM

ESP32 dual-core Xtensa CPU computes Proportional, Integral, and Derivative error corrections for all 3 axes in under 2 ms.
Correction: ΔRoll, ΔPitch, ΔYaw

STAGE 04 — MOTOR MIXER

Equations map corrections into four separate 8-bit to 16-bit PWM duty cycles on GPIO25, 26, 27, and 14.
M_i = Throttle + PID_Roll + PID_Pitch

STAGE 05 — MOSFET DRIVE

A2SHB N-MOSFET switches motor current from the battery rail; 8520 brushed motors modulate RPM immediately, restoring equilibrium.
Response: Instant attitude recovery

MODULE 03.3

COMPONENT MATRIX: ELECTRICAL SUBSYSTEMS & HARDWARE RATINGS

FLIGHT CONTROLLER

ESP32 DevKit V1
3.3V LOGIC
Dual-core 240 MHz MCU managing sensor fusion, attitude estimation, and independent 4-channel PWM generation.
Pinout: GPIO25, 26, 27, 14 (PWM) | GPIO21 (SDA) | GPIO22 (SCL)

POWER SWITCHING

A2SHB N-MOSFET (4x)
LOW-SIDE N-CH
High-speed electronic switch driven by ESP32 PWM signals through a 10kΩ series gate resistor to control motor currents.
Rating: 20V / 2.3A | 10kΩ Gate Pull-down for boot safety

MOTION SENSOR

MPU6050 Gyro + Accel
6-AXIS IMU
Digital motion sensor tracking angular velocity (±2000°/s) and tri-axial acceleration (±16g) via I²C interface.
Protocol: Fast I²C (400 kHz) | Digital Low Pass Filter enabled

BACK-EMF SHIELD

SS34 Schottky Diodes (4x)
SCHOTTKY
Connect each in parallel across each inductive motor coil to safely dissipate inductive flyback voltage spikes during PWM shutoff.
Rating: 40V / 3A | Ultra-fast reverse recovery time

PROPULSION

8520 Brushed DC Motors
4× CORELESS
8.5mm × 20mm micro motors spinning 55mm propellers to produce high thrust-to-weight ratio for responsive hover.
Nominal Voltage: 3.7V - 5.0V | RPM: ~45,000 unloaded

ENERGY STORAGE

Power Rail & 470µF Filter
1S LIPO + 5V BOOST
1S 3.7V LiPo stepped up to regulated 5V via DC-DC boost converter, stabilized by a 470µF 10V bulk electrolytic capacitor.
Battery: 1S LiPo (30C discharge) | Prevents brownouts

ELECTRICAL SAFETY CHECK : COMMON GROUND SHARED ACROSS ESP32, MPU6050, AND MOTOR DRIVER LOW-SIDES ELIMINATES REFERENCE DRIFT AND GROUND LOOPS.

MODULE 04

ESP8266: THE PAYLOAD

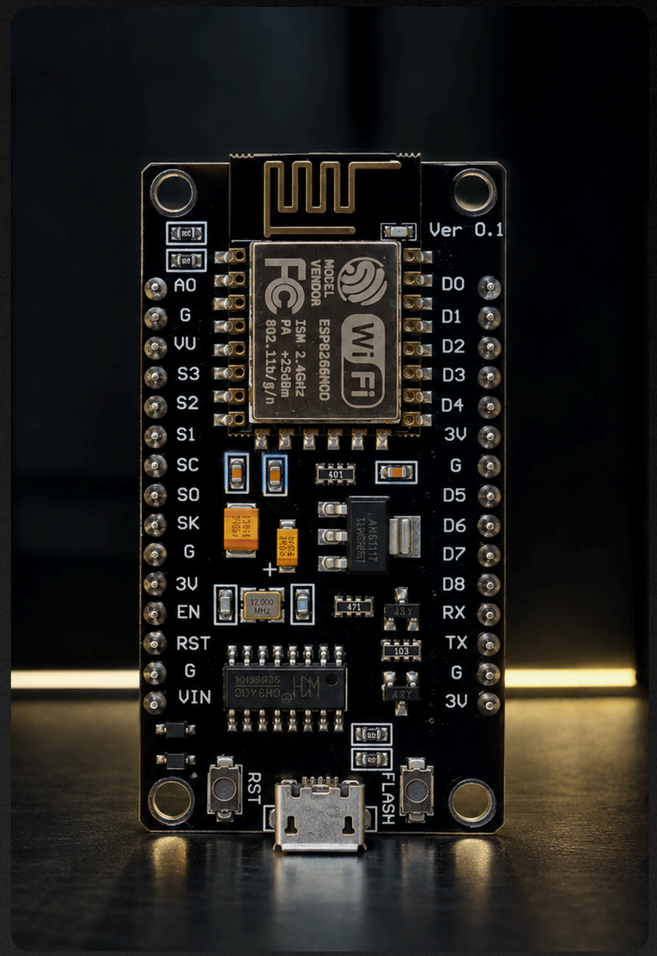
- **SCAN & IDENTIFY:**
  The ESP8266 scans the airwaves to identify active Access Points (BSSID/MAC addresses) and their associated Station clients (MAC addresses).

- **FRAME CRAFTING:**
  It constructs IEEE 802.11 Deauthentication or Disassociation frames, which are management frames telling a connected device to disconnect.

- **SPOOFING SOURCE:**
  It sets the sender MAC address in these frames to match the Access Point's MAC (or vice versa), tricking the client into thinking the AP is kicking it off.

- **BROADCAST INJECTION**:
  It transmits these forged frames on the target channel. Since deauth frames are often sent as broadcasts (to FF:FF:FF:FF:FF:FF) or specific unicast, they bypass standard authentication checks.

- **FORCE RECONNECTION:**
  The client receives the frame, disconnects from the Wi-Fi network, and typically attempts to reconnect immediately, causing a visible disruption or flooding the AP with reconnection requests (DoS effect).



DRONE PROTOTYPE PRESENTATION

Advanced drone technology designed for precision, performance, and limitless aerial possibilities.

a design by aditya - anand